

Stabilizing Prompt-Cache Hit Rates for Claude Code on OpenAI Responses via OpenCodex

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Abstract

This paper measures prompt-cache hit rates for Claude Code and Grok Build on Muse Spark 1.3 contributor via OpenCode Go / OpenCodex, diagnoses Claude's Responses instructions churn, and evaluates OCXFIX. Baseline: Claude mean 0.168384 vs Grok 0.966526 (n=75 each). WHY suite rejects session affinity; raw replay reaches ≈ 0.99 ; conversion shows instructions_sha unique 10/10 from <total_tokens>/TaskCreate appends. Freeze proof lifts S/T4 harness to 0.9934. OCXFIX (FREEZE=0) pilot harness 0.9876 / upstream 0.8892; suite Claude 0.864374 (S/T4 0.982700), Grok 0.941345. Median latency 72.2 s vs 51.8 s; ≈ 48 vs ≈ 97 tok/s.

Keywords: prompt caching, Claude Code, OpenCodex, OpenAI Responses, cache hit rate, coding agents

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Coding-agent harnesses wrapping the same model can still differ in prompt packaging and cache accounting. This study measures that difference, identifies the causal conversion mechanism, and evaluates a product-shaped OpenCodex fix.

Research questions: (1) How large is the baseline hit-rate gap? (2) Session affinity, provider refusal, or request-surface churn? (3) Can inbound OCXFIX recover Claude rates, and at what speed?

Method

Design and Metric

Shared model: Muse Spark 1.3 contributor via OpenCode Go / OpenCodex. Harnesses: Claude Code and Grok Build. Fixtures S/M/L; tasks T1–T5; N=5; 150 cells per suite.

$\text{hit_rate} = \text{cache_read} / (\text{cache_read} + \text{cache_write} + \text{uncached_input})$

Follow-ups: WHY session/upstream/raw-replay; conversion hashes; freeze proxy proof; OCXFIX inbound stabilize (strip <total_tokens>/TaskCreate from instructions, reattach as trailing input; FREEZE=0); speed from suite latency and derived tok/s.

Results

Table 1

Baseline prompt-cache hit rate by harness

Harness	n	Mean	Median	SD	Min	Max
Claude Code	75	0.168384	0.135745	0.075726	0.044133	0.356642
Grok Build	75	0.966526	1.000000	0.069994	0.658178	1.000000

Note. Mean ratio $\approx$ 5.74. Claude S/T4 mean 0.134922.

Figure 1

Baseline hit-rate distributions by harness.

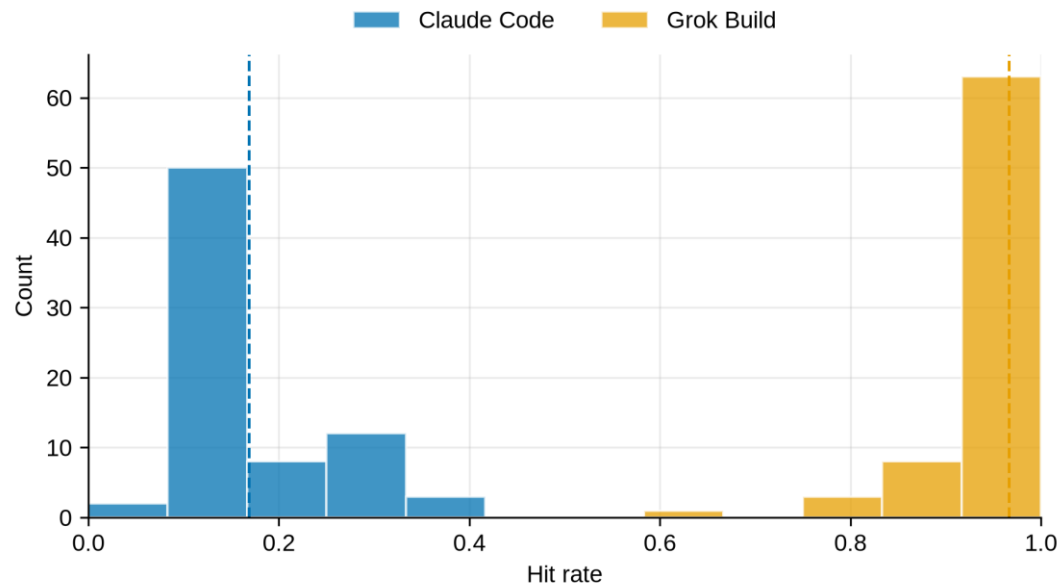
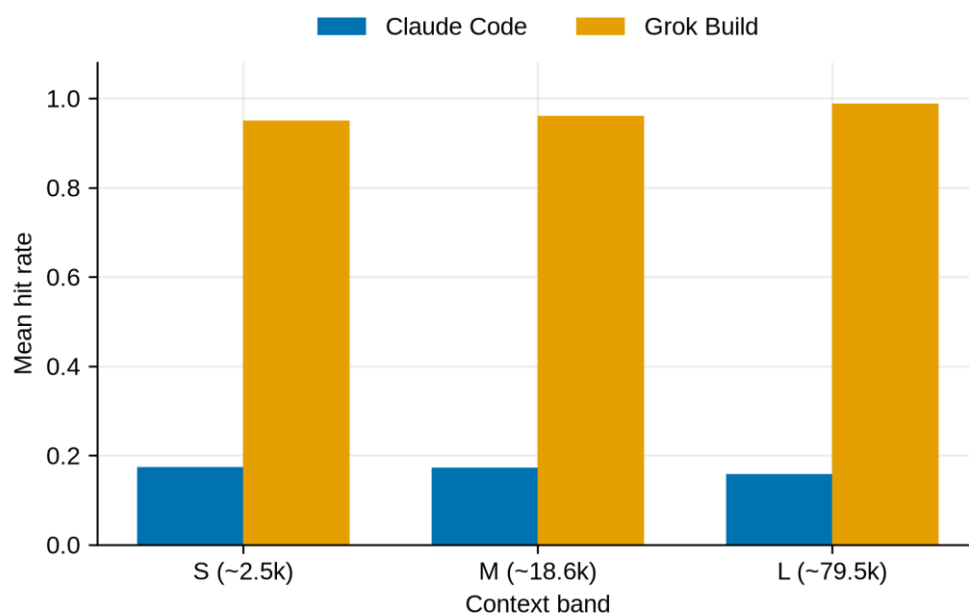


Figure 2

Baseline mean hit rate by context band.

**Figure 3**

Session ablation on band S.

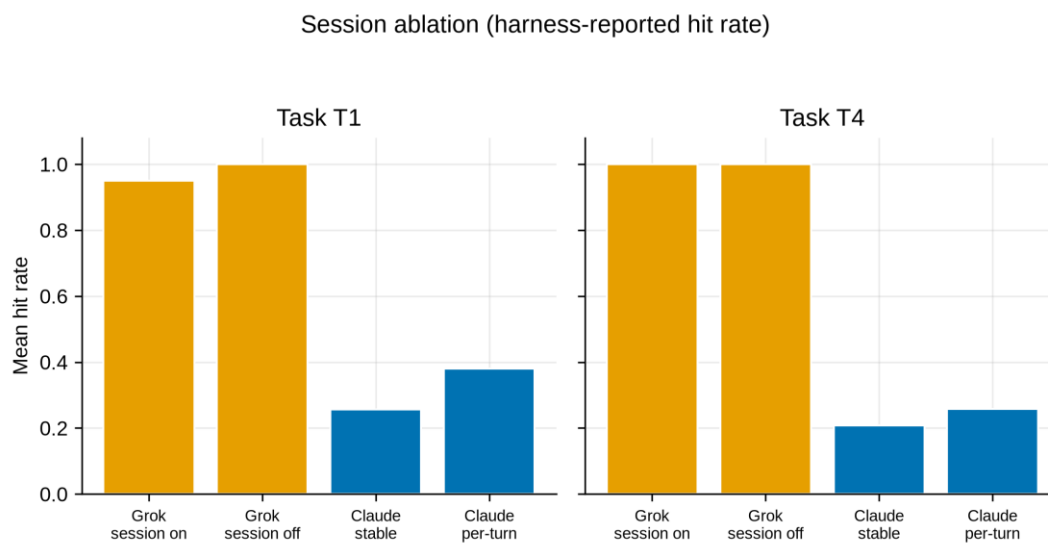
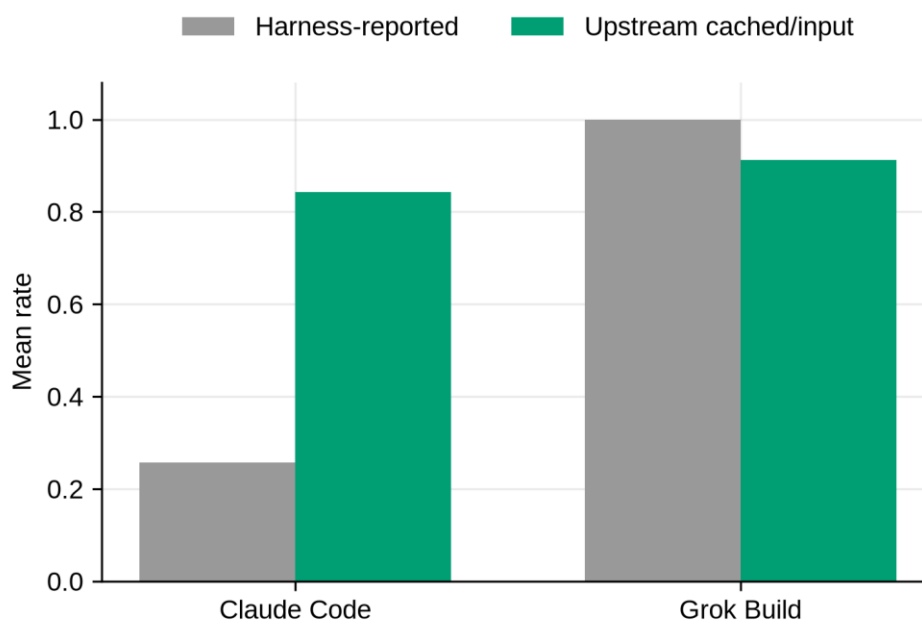
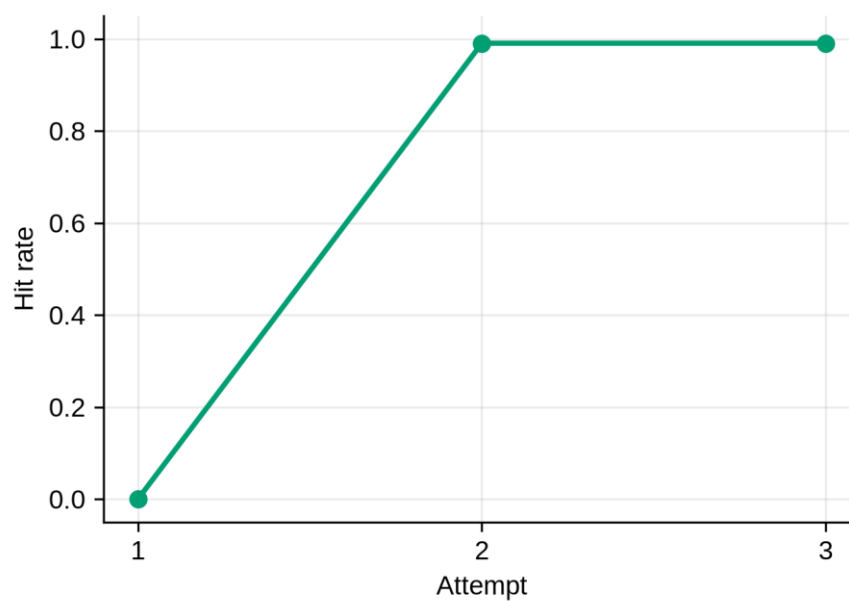


Figure 4

Harness-reported hit rate versus upstream cached/input.

**Figure 5**

Identical raw request replay with a stable prompt cache key.



Conversion: Claude inbound system/tools stable on main turns; outbound instructions_sha unique 10/10; Grok has no instructions field. Diff: appending <total_tokens> footers + rare TaskCreate nudge. Freeze ON S/T4: harness 0.9934; upstream 0.7926 / warm 0.9688 vs conversion ≈ 0.11 .

Figure 6

Claude S/T4 intervention ladder.

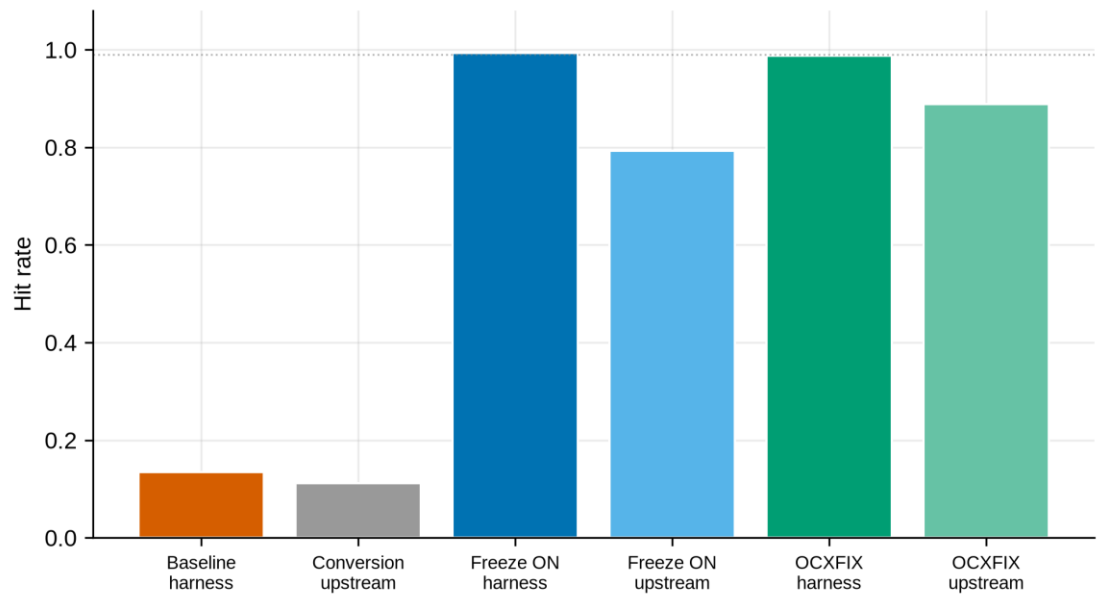


Table 3

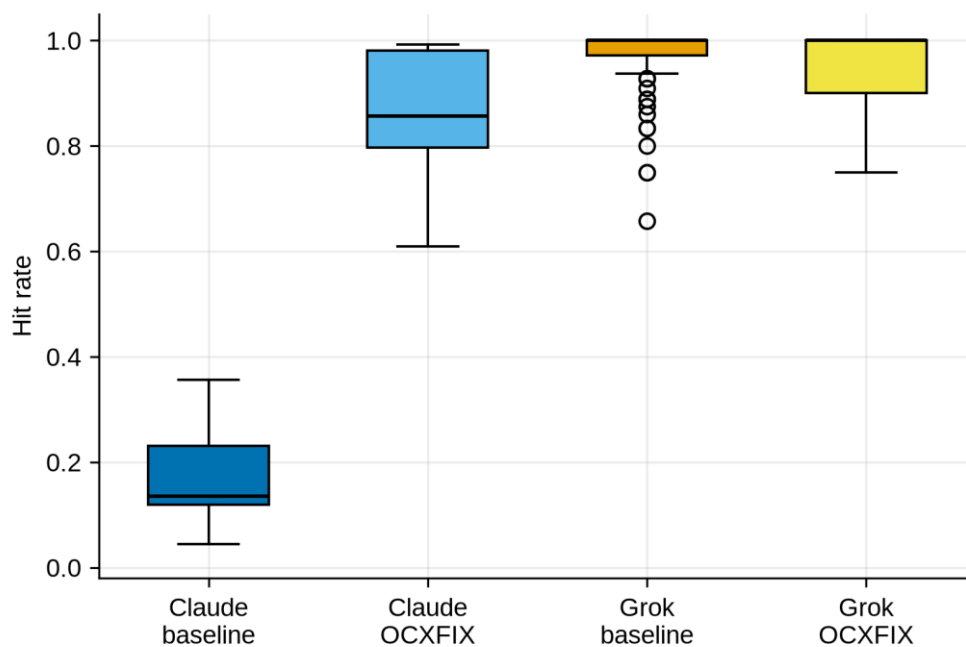
OCXFIX suite versus baseline

Slice	Baseline n	Baseline mean	OCXFIX n	OCXFIX mean	Δ
Claude all	75	0.168384	75	0.864374	0.695990
Grok all	75	0.966526	75	0.941345	-0.025181
Claude S/T4	5	0.134922	5	0.982700	0.847778

Note. OCXFIX pilot S/T4 harness 0.9876; upstream 0.8892 / warm 0.9701.

Figure 7

Hit-rate distributions: baseline versus OCXFIX.

**Figure 8**

Claude mean hit rate by band: baseline versus OCXFIX.

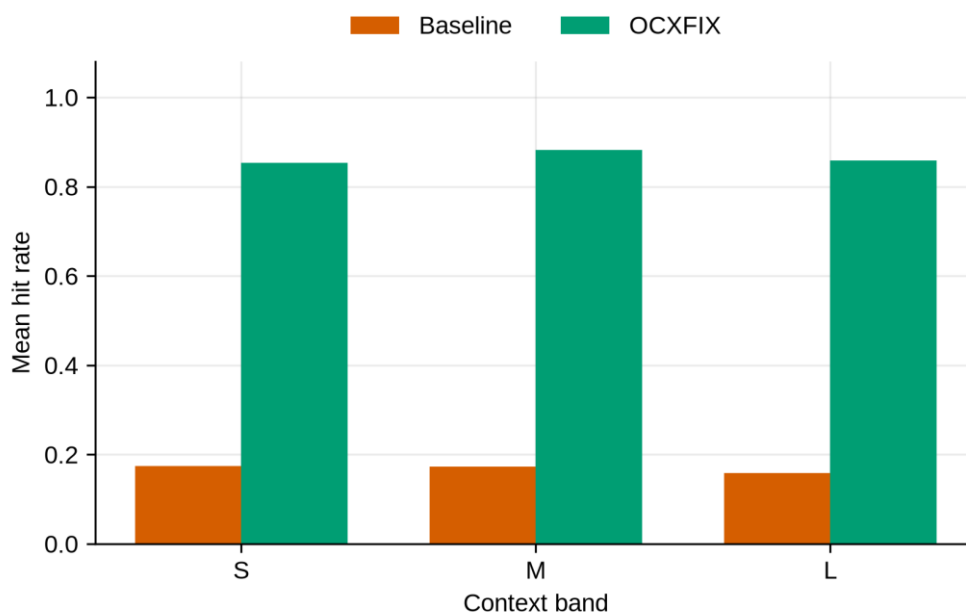
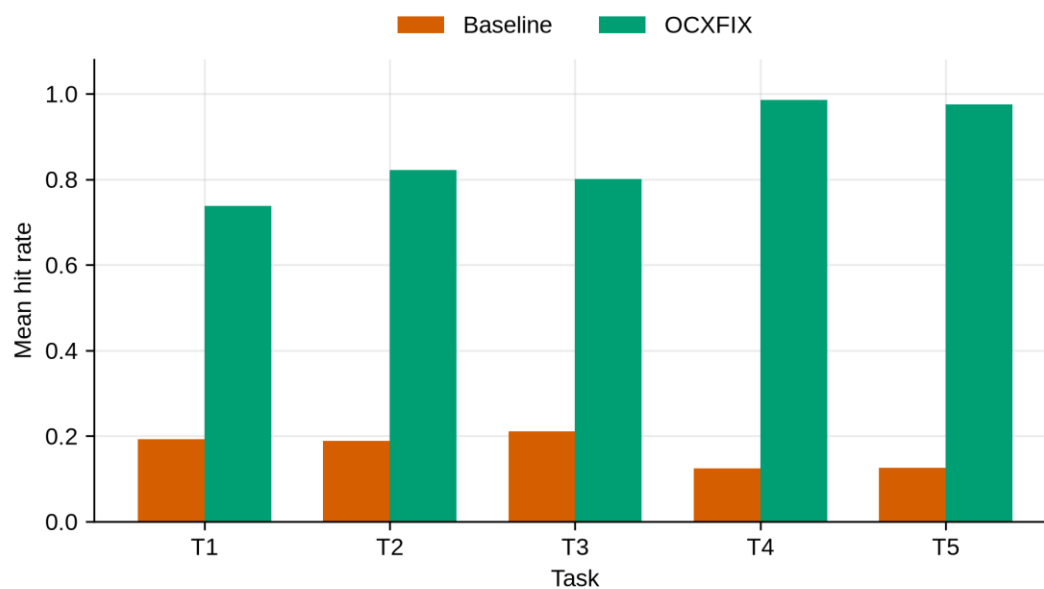


Figure 9

Claude mean hit rate by task: baseline versus OCXFIX.

**Figure 10**

OCXFIX mean hit rate by context band.

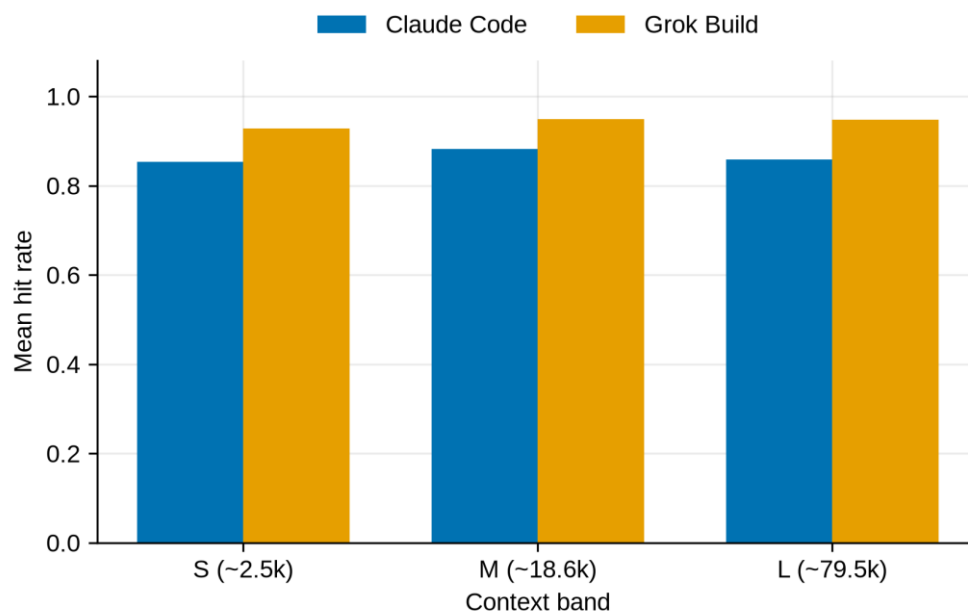


Table 5*OCXFIX median latency and output throughput by band*

Band	Claude n	Claude med ms	Claude med tok/s	Grok n	Grok med ms	Grok med tok/s
S	25	72860.0	52.0	25	38903.0	95.0
M	25	76425.0	47.6	25	70787.0	93.7
L	25	63690.0	46.0	25	51827.0	100.4
All	75	72156.0	47.8	75	51827.0	96.8

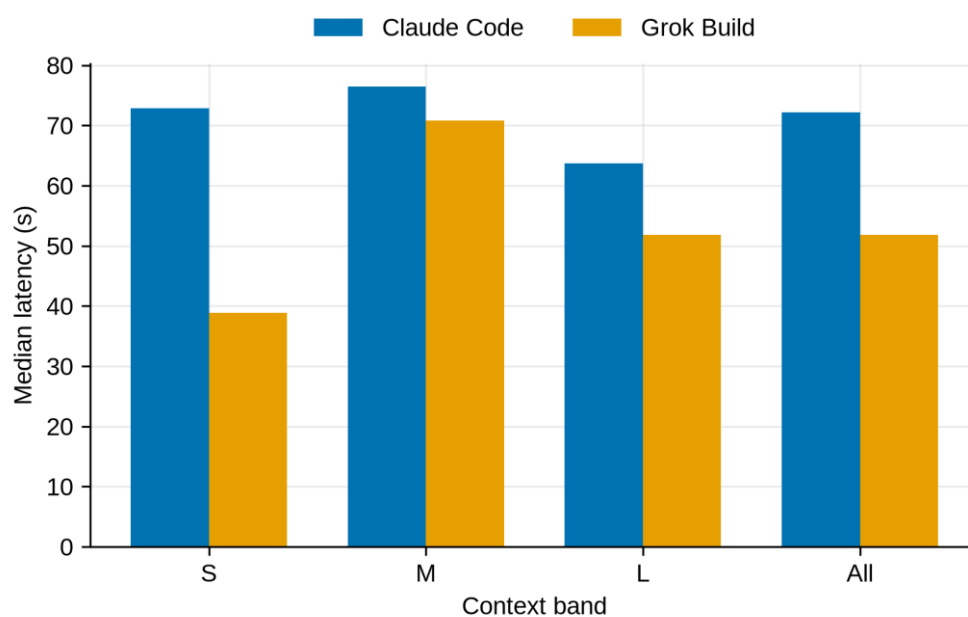
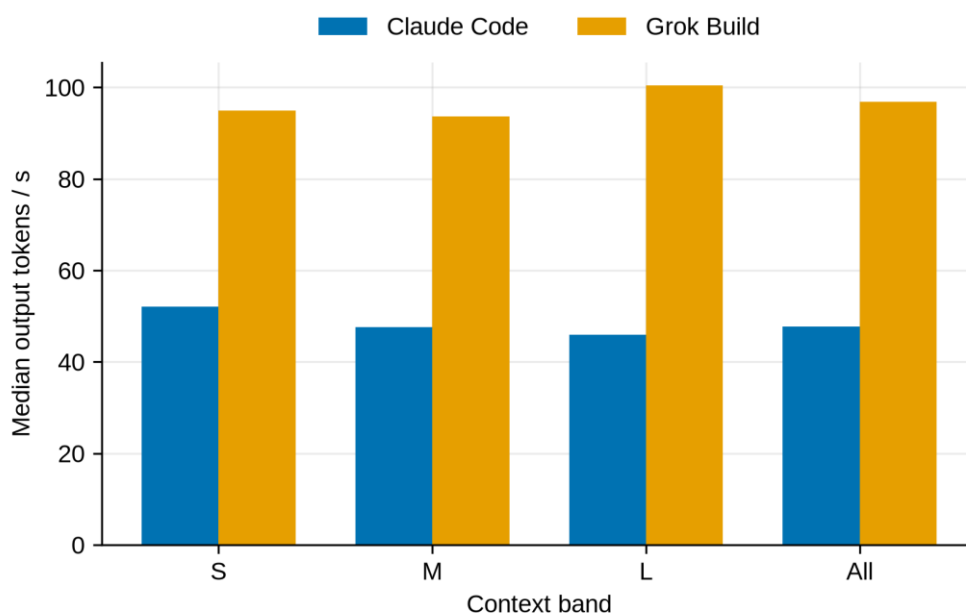
Figure 11*OCXFIX median wall-clock latency by band.*

Figure 12

OCXFIX median output tokens per second by band.



Discussion

Baseline Claude 0.168384 vs Grok 0.966526 ($\approx 5.7\times$) is explained by Responses instructions churn on Anthropic conversion, not session affinity. Freeze proves sufficiency; OCXFIX productizes the fix (suite Claude 0.864374, S/T4 0.982700; Grok 0.941345).

Limitations

Specific to tested harness versions, model route, and fixtures.

Harness/conversion study, not a pure model bake-off.

Upstream alignments approximate; cold first turns expected.

Freeze is proxy-only; product claim is OCXFIX with FREEZE=0. Intermittent rate limits occurred.

Speed is suite wall-clock / derived tok/s, not a TTFT microbenchmark.

Cost not analyzed.

Conclusion

Claude Code's low baseline hit rate (0.168384) versus Grok (0.966526) is caused by outbound instructions churn. OCXFIX raises Claude to 0.864374 (S/T4 0.982700) while Grok stays at 0.941345, with median latency 72.2 s vs 51.8 s and ≈ 48 vs ≈ 97 tok/s.